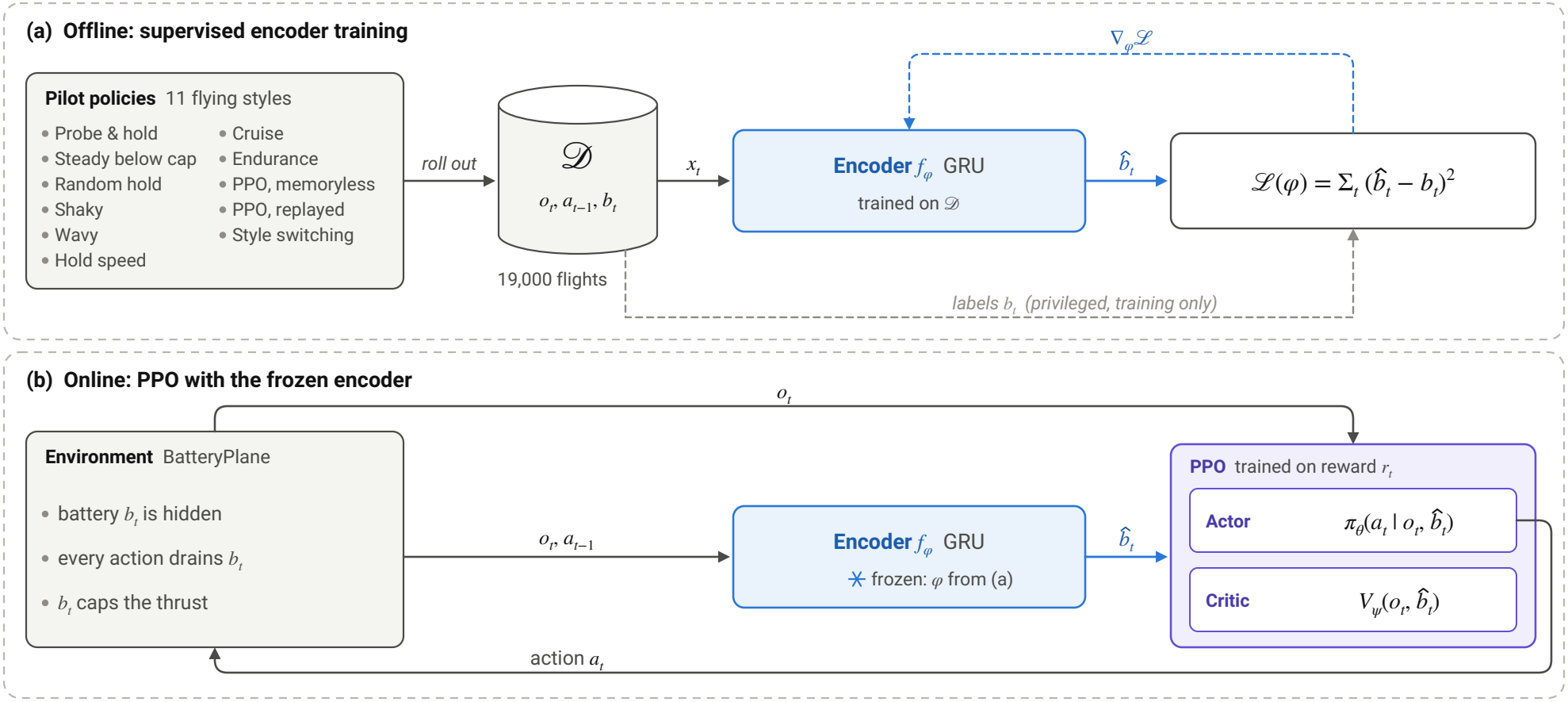

(a) Offline: supervised encoder training
Pilot policies 11 flying styles
Probe & hold
Steady below cap
Random hold
Shaky
Wavy
Hold speed
Cruise
Endurance
PPO, memoryless
PPO, replayed
Style switching
roll out
$\mathcal{D}$
o_t, a_{t-1}, b_t
19,000 flights
x_t
Encoder f_φ GRU
trained on $\mathcal{D}$
$\hat{b}_t$
$\mathcal{L}(\varphi) = \Sigma_t (\hat{b}_t - b_t)^2$
$\nabla_\varphi \mathcal{L}$
labels b_t (privileged, training only)
(b) Online: PPO with the frozen encoder
o_t
Environment BatteryPlane
battery b_t is hidden
every action drains b_t
b_t caps the thrust
o_t, a_{t-1}
Encoder f_φ GRU
frozen: φ from (a)
$\hat{b}_t$
PPO trained on reward r_t
Actor
$\pi_\theta(a_t \mid o_t, \hat{b}_t)$
Critic
$V_\psi(o_t, \hat{b}_t)$
action a_t